

Computational Approach to Neorealism and Constructivism

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Abstract

1. Research Context

The ending of the Cold War introduced an intellectual vacuum as existing paradigms were unable to predict its peaceful conclusion. **Neorealism**, as outlined by Kenneth Waltz [2], theorizes that states act out of self-interest, be it for survival or power accumulation, given the anarchical system that is the world stage.

The USSR undermining their relative gain to the US could be explained through the lens of upcoming **Constructivism**, formalized by Alexander Wendt [3] in 1992 as a direct criticism on Neorealism. Instead of being rigidly self-interested through material constraints, state identities and interests are formed through interaction and are changeable.

As of today, both theories are recognized to have explanatory abilities and even a synthesis was tried [1], though fundamental differences persist.

Agent-based modelling (ABM) is a tried and tested technique within the International Relations (IR) field, but only a single dedicated study seems to exist for constructivist ABM studies. Where RL agents are widely used to model IR ABMs, LLM Agents have little, though rapidly growing research. There seems to be a gap for both dedicated neorealist vs. constructivist research, as well as LLM agent as constructivist agent.

2. Research Methods

2.1. Method Justification

The computational model is a round-based repeated game representing an anarchic system.

General Sufficiency principle - $\hat{z}$ as minimal as possible. How do we produce, with as little elements as possible, the intended macro-behaviour that we want to investigate.

RL is a widely used technique for modelling IR theories, specifically useful for materialist views such as neorealism (as we can optimize our own resources).

LLMs have shown norm copying and identity forming behaviour, similar to the core thesis of Constructivism.

LLMs are actually a good representation of constructivist behaviour.

2.2. Computational Model Design

The **State Space** consists of an array containing the resources of every agent and the history of all recent interactions, conflicts and their outcomes. Each round, all agents choose simultaneously from the **Action Space**:

Action	Target	Effect
Invest	Self	Costs I , generates V (net positive growth)
Invest	Other	Costs I , other receives V (signals trust)
Arm	Self	Costs D , multiplier M/O for 1 round (general threat)
Arm	Other	Costs D , multiplier M/O for 2 rounds (signals alliance)
Attack	Other	Probabilistic conflict resolution (see below)

Conflict Resolution: Attacker vs. Defender (+ allies with active defend). Win probability $P = \frac{\text{attacker power}}{\text{attacker power} + \text{defender power}}$. Winner receives $T\%$ of loser's resources; all combatants pay cost C .

2.3. Agent Architectures

The simulation is done using homogenous agent architectures for every agent.

Reinforcement Learning (RL) Agents represent state actors in Waltz's theory of Neorealism. At its core, the agents should be self-interested and concerned with optimizing their relative gain within the system. Computationally, this identity is translated through optimizing the following function during training: [insert RL optimization function]

Large-Language Model (LLM) Agents represent state actors in Wendt's theory of Constructivism. Based on the intersubjective interpretation of shared history, LLM agents are prompted internally update their identities and interests, and choose their next action accordingly.

Control LLM Agents represent the state actors in Neorealistic theory, but are prompted to optimize their relative gain through their actions. These agents are specifically introduced to effectively isolate the effects of the constructivist prompting.

3. Significance of Research

With the debate between Neorealism and Constructivism still actively ongoing, my research could potentially provide additional insights. Though, ABM models are widely used in IR studies, the Artificial Intelligence approach to designing the agent architecture seems quite novel. Especially, comparative studies between RL and LLM behaviour in social studies is an active field of research.

4. Bibliography

5. Timeline

References

- [1] J. Samuel Barkin. *Realist Constructivism: Rethinking International Relations Theory*. Cambridge University Press, Cambridge, 2010.
- [2] Kenneth N. Waltz. *Theory of International Politics*. Addison-Wesley, Reading, MA, 1979.
- [3] Alexander Wendt. Anarchy is what states make of it: The social construction of power politics. *International Organization*, 46(2):391–425, 1992.